Linear Algebra (MT-1004)

Homework#17

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Question 1

1. Find the projection of $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ onto the subspace spanned by $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$.
2. Determine whether the projection of $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ onto a subspace is unique.
3. Explain the geometric interpretation of projecting onto a subspace.
4. **Python Task:** Implement a function that projects a vector onto the column space of a given matrix.

```
def project_onto_subspace(x, A):
    # Your code here
```